(12) INTERNATIONAL APPLICATION PUBLISHED UNDER THE PATENT COOPERATION TREATY (PCT)

(19) World Intellectual Property Organization
International Bureau

(43) International Publication Date
04 May 2023 (04.05.2023)

WIPO | PCT

(10) International Publication Number
WO 2023/075687 A2

(51) International Patent Classification:
Not classified

(21) International Application Number:
PCT/SG2022/050762

(22) International Filing Date:
25 October 2022 (25.10.2022)

(25) Filing Language: English

(26) Publication Language: English

(30) Priority Data:
10202112046S 29 October 2021 (29.10.2021) SG

(71) Applicant: **NATIONAL UNIVERSITY OF SINGAPORE** [SG/SG]; 21 Lower Kent Ridge Road, Singapore 119077 (SG).

(72) Inventors: **PATWARI, Rishab**; c/o National University of Singapore, Faculty of Engineering / Department of Electrical and Computer Engineering, 21 Lower Kent Ridge Road, Singapore 119077 (SG). **NGUYEN TUAN, Dung**; c/o National University of Singapore, Faculty of Engineering / Department of Mechanical Engineering, 21 Lower Kent Ridge Road, Singapore 119077 (SG). **LIN, Zhicheng**; c/o National University of Singapore, Faculty of Engineering / Department of Mechanical Engineering, 21 Lower Kent Ridge Road, Singapore 119077 (SG). **SEENIVASAN, Praveenkumar**; c/o National University of Singapore, Faculty of Engineering / Department of Electrical and Computer Engineering, 21 Lower Kent Ridge Road, Singapore 119077 (SG).

(74) Agent: **DAVIES COLLISON CAVE ASIA PTE. LTD.**; 10 Collyer Quay #07-01, Ocean Financial Centre, Singapore 049315 (SG).

(81) Designated States *(unless otherwise indicated, for every kind of national protection available)*: AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CV, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IQ, IR, IS, IT, JM, JO, JP, KE, KG, KH, KN, KP, KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, WS, ZA, ZM, ZW.

(84) Designated States *(unless otherwise indicated, for every kind of regional protection available)*: ARIPO (BW, CV, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ, TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, ME, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

Declarations under Rule 4.17:
— *of inventorship (Rule 4.17(iv))*

(54) Title: ROBOT ALIGNMENT AND MANIPULATION

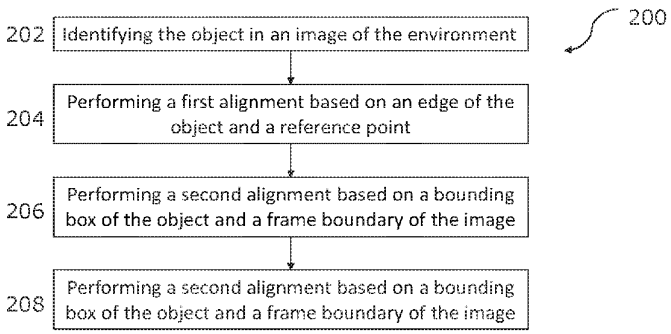


Figure 2

(57) Abstract: Disclosed is a method for aligning an autonomous cleaning robot with an object in an environment. The method includes identifying the object in an image of the environment, performing a first alignment of the autonomous cleaning robot with the object based on an edge of the object and a reference point, performing a second alignment of the autonomous cleaning robot with the object based on a bounding box of the object and a frame boundary of the image, and performing a third alignment of the autonomous cleaning robot based on a distance between the autonomous cleaning robot and the object.

WO 2023/075687 A2

[Continued on next page]

Published:

— *without international search report and to be republished upon receipt of that report (Rule 48.2(g))*

Robot Alignment And Manipulation

Technical Field

The present invention relates, in general terms, to systems and methods for aligning an autonomous cleaning robot with an object, manipulating a robotic arm of that robot, and interchanging tool heads of an end effector.

Background

While the traditional manual method is effective for toilet cleaning, it is labour intensive and time consuming. Moreover, its cleaning effectiveness decreases as the cleaning fluid becomes contaminated. A need exists in the art for an improved method for cleaning a surface to provide an affordable cleaning device for automating wet floor cleaning in the home.

In many large buildings, such as hospitals, large retail stores, cafeterias, and the like, there is a need to clean on a daily or nightly basis, and attempts have been made to address this problem by the development of industrial floor cleaning "robots".

It has become increasingly advantageous to automate certain routine processes, particularly those that are manual labour intensive. Processes involving manual labour can often be difficult to automate on the basis that they require high consistency and low tolerance for error.

It would be desirable to overcome or ameliorate at least one of the above-described problems, or at least to provide a useful alternative.

Summary

Disclosed is a method for aligning an autonomous cleaning robot with an object in an environment, the method comprising:

identifying the object in an image of the environment;

performing a first alignment of the autonomous cleaning robot with the object based on an edge of the object and a reference point;

performing a second alignment of the autonomous cleaning robot with

the object based on a bounding box of the object and a frame boundary of the image; and

performing a third alignment of the autonomous cleaning robot based on a distance between the autonomous cleaning robot and the object.

Performing the first alignment may comprise:

identifying two points defining the edge; and

comparing angles of the two points, each angle being defined between the edge and a line from the respective said point to the reference point. Performing the first alignment may comprise determining a distance between the autonomous cleaning robot and each of the two points.

Performing the second alignment may comprise detecting alignment between the bounding box and frame boundary. Detecting alignment may comprise determining a centre of the bounding box relative to centre region of the image.

Performing the third alignment may comprise:

determining a distance between the autonomous cleaning robot and the edge; and

adjusting a position of the autonomous cleaning robot if the distance is outside a reference range.

The image may be captured by a camera system.

Also disclosed is a method for vision guided manipulation of a robotic arm of an autonomous cleaning robot, comprising:

generating a 3-dimensional (3D) model of an object based on a point cloud of the object;

modelling a cleaning surface of the object, based on an edge of the 3-D model;

forming an array of waypoints along the cleaning surface; and

moving an end effector of the robotic arm between successive ones of the waypoints, to clean the cleaning surface.

Moving the end effector may comprise determining movements of the end effector based on a model of the end effector and the cleaning surface.

Forming an array of waypoints may comprise:

sampling the point cloud at the cleaning surface to produce a point cloud for each waypoint; and

storing the point cloud for each waypoint in memory.

The point cloud may be generated from a depth image captured by a camera system.

Also disclosed is a system for controlling an autonomous cleaning robot, comprising:

memory;

at least one processor (hereinafter **the Processor**); and

a camera system,

wherein the memory stores instructions that, when executed by the Processor, cause the system to perform the method described above.

Also disclosed is an autonomous cleaning robot comprising the system described above.

Also disclosed is an autonomous cleaning robot comprising:

a robotic arm;

a tool head dispenser for dispensing tool heads; and

a tool head disposal unit,

wherein the robotic arm is configured to use a tool head by:

removing the tool head from the tool head dispenser;

using the tool head to clean a surface; and

delivering the tool head to the tool head disposal unit after cleaning the surface.

The robot may further comprise at least one tool holder holding a respective tool to which the tool head is attachable, the tool and tool head together forming an end effector, the robotic arm being configured to connect to a said tool such that using the tool head comprises controlling the tool. The robotic arm and tool may couple using a pneumatic coupling.

The tool head disposal unit may comprise a tool head remover and delivering the tool head comprises removing the tool head using the tool head remover. The tool head remover may align the tool head with a disposal bin.

The robot may further comprise one or more utilities lines extending from a body of the robot to a distal end of the robotic arm. The one or more utilities lines may connect from the distal end to the tool head.

The robot may further comprise a control system configured to perform at least one of:

aligning the autonomous cleaning robot with an object according to the method described above; and

manipulating the robotic arm according to the method described above.

Advantageously, some embodiments provide autonomous computer vision guided robot alignment. The robotic arm automatically adjusts to best fit intended trajectory along a surface in a real environment. This eliminates need to pre-train the robot and calibrate it for fixed use case, hence increasing its versatility.

Advantageously, some embodiments use real-time surface normal estimation – e.g. those that use point cloud, where a surface normal for each point in the cloud is used in surface normal estimation. This enables the robot arm to maintain constant consistent force and distance when manipulating along a surface. For example, a distance from the surface can be calculated and the robot arm thereby controlled (i.e. manipulated) to maintain a consistent distance from the cleaning surface.

Advantageously, some embodiments provide an autonomous tool switching mechanism. These embodiments are able to switch different tools on robot without human intervention.

Brief description of the drawings

Embodiments of the present invention will now be described, by way of non-limiting example, with reference to the drawings in which:

Figure 1 is a robotic arm with depth camera in observe pose;

Figure 2 illustrates an example method for aligning an autonomous cleaning robot with an object in an environment;

Figure 3 shows a bounding box detection and labelling of toilet scene with Yolo5;

Figure 4 shows a toilet bowl image extracted from original image;

Figure 5 shows edges of toilet bowl are detected and alignment w.r.t. feature is calculated;

Figure 6 illustrates image segmentation of toilet bowl into sub-features;

Figure 7A-7C show edges of water cabinet detected and used for angle estimation;

Figure 8 illustrates an example method of performing the first alignment;

Figure 9 illustrates horizontal edges of water cabinet used for distance comparison;

Figure 10 shows a depth camera with isosceles triangle;

Figure 11 is an illustration of origin within alignment margin;

Figure 12 illustrates an example method of performing the third alignment.

Figure 13 shows an example method 1300 for vision guided manipulation of a robotic arm of an autonomous cleaning robot;

Figure 14 shows progressive steps in the identification of a cleaning surface and cleaning of that surface;

Figure 15 relates to a block diagram showing an exemplary computer device, in which embodiments of the invention may be practiced;

Figure 16 shows a display of the splash screen of an autonomous or semi-autonomous robot;

Figure 17 shows a display of the home screen of an autonomous or semi-autonomous robot;

Figure 18 shows a display depicting misalignment of an autonomous or semi-autonomous robot relative to an object (toilet);

Figure 19 shows a display depicting proper alignment of an autonomous or semi-autonomous robot relative to an object (toilet);

Figure 20 shows a display of a cleaning selection screen of an autonomous or semi-autonomous robot;

Figure 21 shows a display of a cleaning in progress of an autonomous or semi-autonomous robot;

Figure 22 shows a display of a cleaning completed screen of an autonomous or

semi-autonomous robot;
Figure 23 shows trolley design with cleaning tool management system;
Figure 24 depicts a tool changer pneumatic base plates;
Figure 25 is an exploded drawing of toilet brush;
Figure 26A-26E show some tools and their customisations;
Figure 27 shows an electrical power delivery sequence;
Figure 28 illustrates a utilities supply tool;
Figure 29 shows tools stored horizontally on magnetic gripper;
Figure 30 illustrates disposable toilet brush tool storage;
Figure 31 is a robot arm attaching toilet brush to toilet brush tool stick;
Figure 32 illustrates toilet brush head disposal;
Figure 33 shows an example point cloud surface normal estimates shows along arm trajectory;
Figure 34 shows a tool holder with passive swapping and disposable tool dispersal system; and
Figure 35 is a robotic arm tool switching mechanism.

Detailed description

The present disclosure relates to autonomous robots with robotic arms. The robots and arms disclosed herein can reduce human involvement in repeated menial tasks (i.e., cleaning toilet bowl, mirrors and welding) and automate tasks requiring multiple tools and surface contact (such as cleaning, welding, manufacturing). In general, the proposed robots and arms can occupy an important position in people's lives because they can automatically do repetitive tasks such as cleaning objects with limited need of manual participation.

In order for the proposed robots (such as cleaning robots) to execute pre-defined interactions with scene objects (for example as shown in Figure 1, the robotic arm 102 manipulates along surface of toilet bowl 106 to perform cleaning task), it is essential that the alignment of the robot to the scene object is similar to the alignment when the trajectory path is trained. As such, a system to determine robot alignment needs to be developed. In the present disclosure, a method for autonomous computer vision guided robot arm trajectory alignment is proposed. Said method allows the robotic arm to align to target cleaning feature (such as toilet bowl, urinal, sink and mirror). In some

embodiments as illustrated in Figure 1, the robotic arm 102 with a camera 104 mounted on its end-effector captures scene from an observe pose 106. This captured scene is then processed for robotic movement control.

As will be discussed in detail, the proposed method involves recognition of the target feature, followed by determination of angle and distance from the feature. The proposed method is an improvement of the traditional YOLO classification algorithm. In particular, the output class list is updated to be just the target features (i.e., toilet bowl, urinal, sink, mirror, shower glass panel) in the toilet to limit the classification model to only relevant features.

Figure 2 illustrates an example method 200 for aligning an autonomous cleaning robot with an object in an environment. The method 200 comprises:

Step 202: identifying the object in an image of the environment;

Step 204: performing a first alignment of the autonomous cleaning robot with the object based on an edge of the object and a reference point;

Step 206: performing a second alignment of the autonomous cleaning robot with the object based on a bounding box of the object and a frame boundary of the image; and

Step 208: performing a third alignment of the autonomous cleaning robot based on a distance between the autonomous cleaning robot and the object.

At step 202, a suitable target object in the scene can be detected (such as toilet bowl 302, potted plant 304 and cup 306) using bounding box classification per Figure 3. In particular, an optimized YOLOv5 object classification algorithm can be used to detect and classify the features (for example the toilet features, including the toilet bowl feature, urinal feature, sink feature and mirror feature) in the scene. YOLOv5 is a family of compound-scaled object detection models trained on the COCO dataset, and includes simple functionality for test time augmentation and hyper parameter evolution. It is an object detection model that combines bounding box prediction and object classification into a single end to end differentiable network.

The target feature (for example the toilet bowl feature, urinal feature, sink feature and mirror feature) is then extracted from the target image obtained. In one example shown in Figure 4, the toilet bowl image (i.e., feature) 402 can

be extracted from the original image 404. It will be appreciated that fixed feature points on the target image (for example rear edges of toilet bowl 402) can then be identified per Figure 5.

After the fixed feature points on the target image are identified, the pre-trained YOLOv5 classification models will then be used to detect key sub-features of common toilet features for classification. Figure 6 illustrates images segmentation of toilet bowl feature into sub-features. As shown in Figure 6, the toilet bowl feature 600 includes water cabinet sub-feature 602 (including flush sub-feature and cabinet sub-feature), toilet bowl base sub-feature 604 (including inner surface and outer surface sub-features), toilet lid sub-feature 606 (including toilet seat cover sub-feature and toilet seat sub-feature). In some embodiments, the urinal feature includes flush sub-feature and urinal container sub-feature. The sink feature includes tap sub-feature, sink surface sub-feature (including inner surface outer surface sub-features). The mirror feature includes mirror surface sub-feature.

After the key sub-features are detected, edge-based image segmentation is run on each feature to identify its edges and classify different segments as shown in Figure 7A. Figure 7B shows the water cabinet detected. In the present disclosure, standard straight edges are identified that can be used for alignment to particular feature. As shown in Figure 7C, two horizontal edges (i.e., P1-P2 and P3-P4) on the water cabinet are detected. Those two edges, as will be discussed in detail, can be used for distance comparison. Table I shows some examples of edges used for different features (not exhaustive).

Toilet Feature	Straight Edges Used
Toilet Bowl	Horizontal edge on wall behind toilet bowl; Horizontal edges on water cabinet;
Urinal	Vertical edges along the side of urinal; Horizontal edges at the top and bottom of urinal;
Sink	Horizontal edges along sink cabinet;

	Vertical edges along the side of sink cabinet;
Mirror	For rectangular mirrors, horizontal & vertical edges along the side of mirror; For circular mirrors, points along arc of mirror edge.

Table I

As illustrated in Figure 8, at step 204, in some embodiments, performing the first alignment comprises:

Step 802: identifying two points defining the edge; and

Step 804: comparing angles of the two points, each angle being defined between the edge and a line from the respective said point to the reference point.

At step 802, as illustrated in Figure 9, two points (i.e., P1 and P2) on the straight edge P1-P2 are chosen and their distance from the reference point 902 is measured using a depth camera. The distance between P1 and point 902 is marked as l1, and the distance between P2 and point 902 is marked as l2.

At step 804, as shown in Figure 10, X1 refers to the angle defined between the edge formed between points 1002 and 1004 and a line from point 1002 to the reference point 1006. Similarly, X2 refers to the angle defined between the edge formed between points 1002 and 1004 and a line from point 1004 to the reference point 1006. In this example, the depth camera is located at the reference point 1006. In some embodiments, performing the first alignment comprises determining a distance between the autonomous cleaning robot (which may also be located at the reference point 1006) and each of the two points 1002 and 1004. If the distances between the two points 1002 and 1004 and the reference point 1006 are similar, the cleaning robot is deemed to be facing the target feature and it is deemed to be normal to the target feature.

At step 206, as shown in Figure 11, performing the second alignment may comprise detecting alignment between the bounding box 1102 and frame boundary of the toilet water cabinet to align the robot horizontally to the centre of the feature. Detecting alignment comprises determining a centre 1104 of the

bounding box 1102 relative to centre region (see 1106) of the image. Said image can be captured by a camera system (for example the depth camera). In particular, an origin 1104 is marked in the centre of the bounding box 1102 of a feature and compared against the centre line (see 1108 and 1110) of the image frame. To be specific, if the image origin lies outside of the alignment margin in the centre of the image frame, it is determined to be not aligned. On the contrary, if the image origin lies within the alignment margin in the centre of the image frame it is determined to be aligned.

As illustrated in Figure 12, at step 206, in some embodiments, performing the third alignment comprises:

Step 1202: determining a distance between the autonomous cleaning robot and the edge; and

Step 1204: adjusting a position of the autonomous cleaning robot if the distance is outside a reference range.

In particular, once the cleaning robot is aligned to be normal to the feature as well as horizontally aligned to the centre of the feature, the pre-set expected distance range from the feature origin is compared against the distance obtained. This pre-set distance is calculated for the different features in order to allow maximum freedom for the arm motion and to reduce chances of collision. If the current distance is greater than expected range, the robot is deemed to be too far away from feature. On the contrary, if the current distance is less than the expected range, the robot is deemed too near to the feature.

The present disclosure also relates to a method for vision guided autonomous robotic Arm manipulation. The proposed method allows autonomous robotic arm manipulation for cleaning on selected target features. It allows detecting surface to be cleaned and stores point cloud map of surface in memory. A data model is used to store the cleaning status of each point on the surface. The proposed method is also able to select points along a surface to form trajectory for arm of the cleaning robot to move and clean. Figure 33 illustrates an example point cloud surface normal estimates shows along arm trajectory.

Figure 13 illustrates an example method 1300 for vision guided manipulation of a robotic arm of an autonomous cleaning robot, comprising:

Step 1302: generating a 3-dimensional (3D) model of an object based on a point cloud of the object;

Step 1304: modelling a cleaning surface of the object, based on an edge of the 3-D model;

Step 1306: forming an array of waypoints along the cleaning surface; and

Step 1308: moving an end effector of the robotic arm between successive ones of the waypoints, to clean the cleaning surface.

At step 1302, in an observe pose, the depth camera mounted on the robot arm can capture the image of the toilet feature. The point cloud from the depth camera of the scene can be used to generate a 3D model that is represented in the MoveIt Planning Scene, which allows one to setup virtual environments (i.e., scenes), create start and goal states for the robot interactively, test various motion planners, and visualize the output.

At step 1304, edge based image segmentation is done to identify and detect the surface of the feature. Table I above shows examples of the edges that are analysed for the feature detection.

Figure 14 shows the progressive steps 1402 to 1407 in identification of shapes, one of which is a cleaning surface, and the cleaning of that surface. Steps 1402 and 1403 relate to modelling a cleaning surface of the object, based on an edge of the 3-D model (i.e., step 1304 of method 1300). In particular, step 1402 relates to edge detection on toilet feature to identify relevant edges (i.e., outer ellipse 1408 and inner ellipse 1409). Step 1403 relates to identifying cleaning surfaces (i.e., 1410) and toilet seat surface (i.e., 1411) bounded by edges 1408 and 1409.

Steps 1404-1407 relate to the process of forming an array of waypoints along the cleaning surface (i.e., step 1306 of method 1300). Step 1404 relates to forming suitable shape used to mark waypoints path (see 1412) along edge 1409. Step 1405 illustrates ellipse of Toilet Bowl with origin marked waypoints. As shown from steps 1405 to 1407, as the cleaning process is going on, the area of the unclean surface will be decreased. The new edge of the unclean surface will be updated as the area of the unclean surface is decreasing. The

new waypoint path 1412 will be updated accordingly. Finally, as shown at step 1407, the trajectory is terminated when cleaning surface has been fully covered.

The present disclosure now discloses details regarding the method of generating path trajectory. First, a suitable shape based on the edge is used to model the cleaning surface of the toilet feature. (i.e., ellipse 1409 for inner toilet bowl). Second, the Cartesian points of all the point clouds points located within the shape is stored in an array of tuples of the form (cartesian_coordinate, pass_count), where pass_count is initialised to 0. The centre of the shape detected in step 1404 is identified as the origin located at Cartesian point (xo, yo, zo). Third, forming an array of waypoints comprises sampling the point cloud at the cleaning surface to produce a point cloud for each waypoint; and storing the point cloud for each waypoint in memory. Said point cloud may be generated from a depth image captured by a camera system. In particular, point clouds lying along the shape where zi = zo are sampled ((xi,yi,zi), pass_count) stored in an array denoting the waypoint for the arm cleaning trajectory. Said array can be formalized as:

waypoints = [((x1,y1,zo), 0), ((x2,y2,zo), 0), …].

In some embodiments, moving the end effector comprises determining movements of the end effector based on a model of the end effector and the cleaning surface It will be appreciated that when a robotic arm with the appropriate end-effector (such as cleaning tool head) would move along the path denoted by the waypoints 1412 in steps 1404 to 1407, the end-effector (such as toilet brush) would contact the points on the surface of the toilet feature within the radius of the end-effector. The 3D planning scene mentioned in step 1302 of method 1300 would be aware of the 3D model of the end-effector as it would be pre-loaded and the position of the end-effector would be used as targets for the inverse kinematics planner to actuate the robot arm. In the present disclosure, the inverse kinematic planner used for testing is the OMPL library. Hence, after the first set of waypoints are stored for zi = zo, the z-coordinate is decremented such that zi = z0 - (radius of end effector). Step 1308 of method 1300 are then repeated until the termination condition which is when zi exceeds the depth of the 3D model or if the position would be beyond the robot arm's reach.

After the entire trajectory for the cleaning sequence has been identified, the path is simulated in MoveIt against the planning scene which contains the detected obstacle. In addition, if all the points do not lead to any collision, the path is accepted and can be used for cleaning. During the actual path execution, the joint angles of the robot are continuously monitored, if there is a deviation of more than 4 degrees for the joint angle on any axis as compared to the expected angle, the trajectory is halted and the operator is alerted.

The present disclosure also relates to system for controlling an autonomous cleaning robot, comprising:

memory;

at least one processor (hereinafter **the Processor**); and

a camera system,

wherein the memory stores instructions that, when executed by the Processor, cause the system to perform the method 200 and 1300. The present invention also relates to an autonomous cleaning robot comprising the proposed system.

Figure 15 is a block diagram showing an exemplary system (such as computer device) 1500, in which embodiments of the invention may be practiced. The computer device 1000 may be a mobile computer device such as a smart phone, a wearable device, a palm-top computer, and multimedia Internet enabled cellular telephones when used in training the model, and, for use in controlling a vehicle or other machine for autonomous driving, may be an on-board computing system or a mobile device such as an iPhone TM manufactured by AppleTM, Inc or one manufactured by LGTM, HTCTM and SamsungTM, for example, or other device in communication with the vehicle or other machine and configured to send control commands thereto and to receive information on human interventions from the vehicle or other machine.

As shown, the mobile computer device 1500 includes the following components in electronic communication via a bus 1506:

(a) a display 1502;

(b) non-volatile (non-transitory) memory 1504;

(c) random access memory ("RAM") 1508;

(d) N processing components 1510;

(e) a transceiver component 1512 that includes N transceivers; and

(f) user controls 1514.

Although the components depicted in Figure 15 represent physical components, Figure 15 is not intended to be a hardware diagram. Thus, many of the components depicted in Figure 15 may be realized by common constructs or distributed among additional physical components. Moreover, it is certainly contemplated that other existing and yet-to-be developed physical components and architectures may be utilized to implement the functional components described with reference to Figure 15.

The display 1502 generally operates to provide a presentation of content to a user, and may be realized by any of a variety of displays (e.g., CRT, LCD, HDMI, micro-projector and OLED displays).

In general, the non-volatile data storage 1504 (also referred to as non-volatile memory) functions to store (e.g., persistently store) data and executable code. The system architecture may be implemented in memory 1504, or by instructions stored in memory 1504.

In some embodiments, the non-volatile memory 1504 includes bootloader code, modem software, operating system code, file system code, and code to facilitate the implementation components, well known to those of ordinary skill in the art, which are not depicted nor described for simplicity.

In many implementations, the non-volatile memory 1504 is realized by flash memory (e.g., NAND or ONENAND memory), but it is certainly contemplated that other memory types may be utilized as well. Although it may be possible to execute the code from the non-volatile memory 1504, the executable code in the non-volatile memory 1504 is typically loaded into RAM 1508 and executed by one or more of the N processing components 1510.

The N processing components 1510 in connection with RAM 1508 generally operate to execute the instructions stored in non-volatile memory 1504. As one of ordinarily skill in the art will appreciate, the N processing components 1510 may include a video processor, modem processor, DSP, graphics processing unit (GPU), and other processing components.

The transceiver component 1512 includes N transceiver chains, which may be used for communicating with external devices via wireless networks. Each of the N transceiver chains may represent a transceiver associated with a particular communication scheme. For example, each transceiver may correspond to protocols that are specific to local area networks, cellular networks (e.g., a CDMA network, a GPRS network, a UMTS networks), and other types of communication networks.

The system 1500 of Figure 15 may be connected to any appliance 418, such as one or more cameras mounted to other robots or fixed in the building in which the robot is operating, to provide scene information, a clock (e.g. so automatically clean areas at a particular time of day or night) or an external database from which context can be acquired.

It should be recognized that Figure 15 is merely exemplary and in one or more exemplary embodiments, the functions described herein may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored on or transmitted over as one or more instructions or code encoded on a non-transitory computer-readable medium 1504. Non-transitory computer-readable medium 1504 includes both computer storage medium and communication medium including any medium that facilitates transfer of a computer program from one place to another. A storage medium may be any available medium that can be accessed by a computer.

The present disclosure now illustrates human user interfaces (UIs) for controlling the proposed multi-functional cleaning robot. These UIs allow human cleaner to operate semi-automated cleaning robot, where the robot may be automatic in other embodiments. They can also guide human to align robot with toilet feature assisted with the methods 200 and 1300 described above.

Figures 16 to 22 shows some example UIs, and they illustrate the cleaning sequence and the corresponding design to control the robot through the application. Table II provides details of these UIs and their functions.

Related Figure	UI Name	Functions
16	Splash Screen	It shows loader while robot is being turned on and initialized.
17	Home Screen	It is an intuitive screen to allow human to select desired toilet feature to clean.
18	Alignment Screen	It guides robot to align to toilet feature using computer vision methods described above. Namely, the angular alignment, horizontal offset from centre and the distance from the toilet feature are checked.
19	Cleaning Selection Screen	It reminds user to lock robot wheels and requests user input to select cleaning options.
20	Cleaning Screen	It displays the tasks that the robot will perform and its current status as well as time elapsed.
21	Cleaning in progress screen	It shows each of the tasks in the cleaning process, and the state (e.g. not start, in process, completed) of each task.
22	Cleaning completed screen	It shows all the tasks in the cleaning process have been completed.

Table II

It will be appreciated that the design for the user interface had to have the considerations shown in Table III based on user feedback and analysis:

User Need/Behaviour	Consequential Design Choice
The user will control the robot using a touch screen panel mounted on the robot body.	The buttons have to be large and easily identifiable.
The user needs properly align the robot to a toilet feature.	A guiding box is added in the alignment screen to make alignment more intuitive.
User needs to charge battery and replace cleaning consumables when they run out.	Battery and consumable icons and their status is shown at all times throughout operation.

Table III

The present disclosure now discloses a method for multiple cleaning tool management. The proposed method allows switching between multiple cleaning tool heads required for different cleaning processes. It also allows power, control and utilities (such as detergent, water and air) delivery to the cleaning tool, as well as enables proper sanitation and disposal of used cleaning tool.

To illustrate the proposed method for multiple cleaning tool management, this disclosure first describes a cleaning tool modification mechanism. In order to accommodate a multiple cleaning tool setup and their intended cleaning role, a series of cleaning tools had to be modified so that they can stored easily, have a common switching mechanism, and a common process to deliver electrical power, utilities like water, detergent, etc..

Figure 23 shows an example autonomous cleaning robot 2300 with the cleaning tool system features annotated, comprising:

a robotic arm 2302 (e.g. elephant robotics mycobot 320 M5 robotic arm);

a tool head dispenser 2304 for dispensing tool heads; and

a tool head disposal unit 2306,

wherein the robotic arm 2302 is configured to use a tool head (not shown) by:

removing the tool head from the tool head dispenser 2304;

using the tool head to clean a surface; and

delivering the tool head to the tool head disposal unit 2306 after cleaning the surface.

In some embodiments, the robot 2300 further comprises at least one tool holder 2308 holding a respective tool (not shown) to which the tool head is attachable. The tool and tool head together form an end effector. The robotic arm 2302 is configured to connect to a said tool such that using the tool head comprises controlling the tool. The robot 2300 may further comprises a dust bin for containing rubbish generated during the cleaning process.

In some embodiments, the robotic arm 2302 and tool couple using a pneumatic coupling. For each tool, tool switching is done using a set of pneumatic base plates (for example 2400 in Figure 24) that would attach to each other through

air compression from a centrally controlled air compressor. In order to attach a tool, the robot arm 2302 would align the plate on its end-effector to the tool's base plate 2400 and then activate the air compression.

Also disclosed is tool stick customisation. Figure 25 illustrates am exploded drawing of a toilet brush 2500, of which the common features that are present on all tools. Table IV shows a list of tool components of the toilet brush 2500 and their purposes.

Tool Component	Purpose
Robot Arm Connector 2502	The arm connector 2502 is a joint on the robot arm to which custom attachments like pneumatic plates can be attached.
Pneumatic Base Plates 2504	The plates 2504 are main form of switching tools. The base plate connected to the robot arm connector has a pneumatic valve that is controlled by the main system. This allows a plate 2504 to attach to other base plates on the tools systems.
Custom Tool Control Box 2506	The control box 256 is used to control different tools which may require different types of actuation. Some tools require rotation with a DC motor. Some tools require a linear actuator and some nothing at all. The common design is to attach these actuators in between the tool base plate 2504 and tool stick 2508 as shown.
Custom Tool Stick 2508	The stick 2508 acts as the shaft to extend the tool length. For the toilet brush tool 2500, its tool stick 2508 has an end which allows disposable toilet brushes to be attached and detached freely.

Cleaning Attachment 2510 (e.g. Cleaning Brush/ Tool Head)	Different cleaning tasks require different cleaning attachments, hence specific cleaning attachments are identified for various tasks.

Table IV

In particular, Figure 26A-26E list example tools and their customisations. Figure 26A illustrates a toilet brush for toilet bowl or urinal cleaning. The custom tool control box of this toilet brush is for controlling a DC motor type spin toilet brush. The control tool stick of the toilet brush has an end which is able to snap onto toilet brush. The tool head of the toilet brush is a Chlorox disposable head.

Figure 26B illustrates a gripper for grasping items such as flush and toilet bowl lid. The custom tool control box of this gripper is for controlling a stepper motor for rotation. The control tool stick of the gripper is a pneumatic system. The tool head of the gripper is a gripper-type head.

Figure 26C illustrates a duster for dusting. The custom tool control box of this duster is for controlling a DC motor for spin mop. The control tool stick of the duster has an internal water and detergent pipe. The tool head of the duster is a mop-type head.

Figure 26D illustrates a dry sponge for drying surface after cleaning. The control tool stick of the dry sponge has a dry sponge holder fixed to stick end. The tool head of the dry sponge is a replaceable dry sponge.

Figure 26E illustrates a wiper for mirror/glass panel wiping. The custom tool control box of this wiper is for controlling a linear actuator to extend and contract wiper. The control tool stick of the wiper has an end connected to fixed wiper holder. The tool head of the toilet brush is a wiper-type head.

Also disclosed is electrical power delivery and utilities delivery. Electrical power and control wires are required to control the end-effector (such as the cleaning tool). However, since the arm 2302 is not always connected to the cleaning tool, one of the major challenges for designing is how to handle electrical power delivery.

The proposed design shown above consists of the custom tool box which is where the actuators for electrically controlled tools are needed. In order to have a standard approach for all tools, this embodiment used an 8-pin electrical module for tool changers to transfer electrical signal and power between the plates to the tool box. Figure 27 illustrates an example electrical power and signal delivery set-up 2700. In the first step 2702, electrical wires pass through the robot arm. In the second step 2704, said wires are connected to an electrical module on the bases plates 2504. In the third step 2706, pins on said electrical module form electrical joint when attached. In the final step 2708, electrical wires are connected to the tool box 2506 actuating tool head.

For utilities delivery, we create an additional tool shown in Figure 28 that can be switched to just like the tools mentioned. In general, in the present disclosure, the robot 2300 comprises one or more utilities lines extending from a body of the robot 2300 to a distal end of the robotic arm 2302. The one or more utilities lines may connect from the distal end to the tool head. The utility pipe (i.e., line) 2800 is able to transfer fluids like water, water mixed with detergent, and air through its nozzle and be used for applications. The control for the fluid is done directly at source and no additional electrical control box is required for this tool.

For tool storage, the robot body is modified to support the arm in mounting and dismounting different tools. As seen from Figure 29, the tools are stored on a gripper 2904 which holds on to the tools 2902 in a horizontal position. This gripper is designed to be magnetic so it can snap on to the tool and keep it in place as the tool base plates are conductors as well.

For special tools like the toilet brush tool, which is disposable, the present disclosure also provides a mechanism to store the toilet brush refills. The design of the disposable tool storage 3000 is shown as seen in Figure 30. In order to mount a disposable toilet brush using the toilet brush attachment, the robot arm 3102 may simply have to press down the tool stick 3104, guided by the aligned 3106 as shown in figure 31.

A sanitation system is required in order to dispose of used toilet brushes after each toilet cleaning operation. For this, a dustbin attachment 3202 (see Figure 32) is added which can be loaded with a disposable plastic bag by a human cleaner. After each cleaning operation, the robot arm 3204 can align above the toilet brush and snap it off from the toilet brush tool stick. As illustrated in Figure 32, at the first step, the robot arm 3204 pulls off the too stick 3208 into the aligner 3206. At the second step, the arm 3204 pulls off the tool stick 3208 from the used toilet brush in the direction 3210. At the final step, the too stick 3208 will be attached to a new toilet brush stored in the refill storage 3212.

In some embodiments, the robot further comprises a control system configured to perform at least one of:

aligning the autonomous cleaning robot with an object according to the proposed method; and

manipulating the robotic arm according to the proposed method.

In some embodiments, the autonomous robot is aligned as follows. First, a suitable target object in the scene is detected (such as a toilet bowl) using bounding box classification. Second, fixed feature points 3302 on the target image are identified (such as rear edges of toilet bowl) per Figure 33. In particular, A vector 3304 is drawn to the feature 3302 and the displacement and angle of the robotic arm with respect to the feature point 3302 is detected. Third, the current angle and displacement is compared to the pre-defined target angle and displacement and the necessary adjustment to the robot's pose is suggested. When the robot is within an acceptable threshold for angle and displacement, it is determined to be aligned to the object.

In fact, the present disclosure proposes a real time surface normal estimation mechanism. Real time surface normal estimation allows the robot to detect the normal to the surface as it is manipulating and correct its trajectory in order to maintain a constant displacement from the surface. In particular, depth camera is used to generate a visual 3D map of the environment. Point cloud data points are analysed and surface is identified. Using surface normal estimation, the normal to the surface is identified (shown in Figure 33 as arrows 3304). As the robot arm moves along its predefined trajectory, the normal displacement of the arm from the surface is computed and the value is then processed. If the

displacement (x cm) is different from the expected displacement at the points 3302, then the arm adjusts its joint angles to move the arm closer or further away from the surface based on the requirement. It will be appreciated that in this way the robot arm is able to maintain a constant distance with the surface and such design also improves the reliability of the system as issues with alignment of the robot with target feature can be compensated for through the surface normal estimation correction.

In the present disclosure, robotic operating system is used for communication between the following nodes responsible for the corresponding actions. Table V illustrates nodes for which the robotic operating system controls communication.

S/N	Node	Description
1	Camera	The camera of the robotic operating system publishes image from the mounted depth camera at a fixed frequency.
2	Computer Vision Scene Detector	Said detector uses object classification algorithm (such as YOLO) to detect objects in a scene using a bounding box classification and identify the objects of interest (which may be pre-defined).
3	Computer Vision Alignment	It identifies common feature points on objects of interest and calculates angular and physical displacement from the feature points to describe alignment of robot to the object.

Table V

The present disclosure also relates to an autonomous tool switching mechanism illustrated in Figure 35. The purpose of said switching mechanism is that the autonomous robot needs to be able to wield different tools automatically in order to perform multiple tasks through the same platform. This is achieved by designing and implementing the following sub-systems. The first sub-system is a tool holder that is able to allow passive swapping of its end effector (see 3400 of Figure 34). The second sub-system is a disposable tool dispersal system. The third sub-system is an active tool swapping mechanism that allows robotic arm to switch between different tool holders. One pneumatic plate is the end effector of the robotic arm while the other plate is attached to the top of the tool holder.

In order to support hot swapping of different tool holders, a pneumatic system is used at the end effector of the robotic arm. It consists of two base plates (see the third sub-system mentioned above), one of which is attached to the end of the robotic arm and the other is on individual tool holders. When the arm attaches to a tool holder, the pneumatic plates will align and join each other.

The tool-holder in turn has been designed such that it can connect and disconnect with various disposable tools. For example, the tool holder (see the first sub-system above), is able to load a disposable toilet brush from the toilet brush dispenser by aligning and then pushing down onto the brush. A fork finger 3402 on the tool holder 3400 locks on to the inner wall of the disposable toilet brush.

When a tool (such as the toilet brush) is dispensed from the dispenser (see the second sub-system above), the dispenser will activate its linear actuator which will push up the next toilet brush to the top ready to be dispensed next. When a disposable tool like the toilet brush is to be disposed of, the tool holder has a latch which releases the fork finger lock when pressed hence releasing the tool attached. Such design is used for disposal of used tools after they are no longer required.

It will be appreciated that many further modifications and permutations of various aspects of the described embodiments are possible. Accordingly, the described aspects are intended to embrace all such alterations, modifications,

and variations that fall within the spirit and scope of the appended claims or statements.

Throughout this specification and the claims or statements that follow, unless the context requires otherwise, the word "comprise", and variations such as "comprises" and "comprising", will be understood to imply the inclusion of a stated integer or step or group of integers or steps but not the exclusion of any other integer or step or group of integers or steps.

The reference in this specification to any prior publication (or information derived from it), or to any matter which is known, is not, and should not be taken as an acknowledgment or admission or any form of suggestion that that prior publication (or information derived from it) or known matter forms part of the common general knowledge in the field of endeavour to which this specification relates.

Claims

1. A method for aligning an autonomous cleaning robot with an object in an environment, the method comprising:

identifying the object in an image of the environment;

performing a first alignment of the autonomous cleaning robot with the object based on an edge of the object and a reference point;

performing a second alignment of the autonomous cleaning robot with the object based on a bounding box of the object and a frame boundary of the image; and

performing a third alignment of the autonomous cleaning robot based on a distance between the autonomous cleaning robot and the object.

2. The method of claim 1, wherein performing the first alignment comprises:

identifying two points defining the edge; and

comparing angles of the two points, each angle being defined between the edge and a line from the respective said point to the reference point.

3. The method of claim 2, wherein performing the first alignment comprises determining a distance between the autonomous cleaning robot and each of the two points.

4. The method of any one of claims 1 to 3, wherein performing the second alignment comprises detecting alignment between the bounding box and frame boundary.

5. The method of claim 4, wherein detecting alignment comprises determining a centre of the bounding box relative to centre region of the image.

6. The method of claim 1 or 2, wherein performing the third alignment comprises:

determining a distance between the autonomous cleaning robot and the edge; and

adjusting a position of the autonomous cleaning robot if the distance is outside a reference range.

7. The method of any one of claims 1 to 6, wherein the image is captured by a camera system.

8. A method for vision guided manipulation of a robotic arm of an autonomous cleaning robot, comprising:

generating a 3-dimensional (3D) model of an object based on a point cloud of the object;

modelling a cleaning surface of the object, based on an edge of the 3-D model;

forming an array of waypoints along the cleaning surface; and

moving an end effector of the robotic arm between successive ones of the waypoints, to clean the cleaning surface.

9. The method of claim 8, wherein moving the end effector comprises determining movements of the end effector based on a model of the end effector and the cleaning surface.

10. The method of claim 8 or 9, wherein forming an array of waypoints comprises:

sampling the point cloud at the cleaning surface to produce a point cloud for each waypoint; and

storing the point cloud for each waypoint in memory.

11. The method of any one of claims 8 to 10, wherein the point cloud is generated from a depth image captured by a camera system.

12. A system for controlling an autonomous cleaning robot, comprising:

memory;

at least one processor (hereinafter **the Processor**); and

a camera system,

wherein the memory stores instructions that, when executed by the Processor, cause the system to perform the method of claim 7 or 11.

13.An autonomous cleaning robot comprising the system of claim 12.

14.An autonomous cleaning robot comprising:

a robotic arm;

a tool head dispenser for dispensing tool heads; and

a tool head disposal unit,

wherein the robotic arm is configured to use a tool head by:

removing the tool head from the tool head dispenser;

using the tool head to clean a surface; and

delivering the tool head to the tool head disposal unit after cleaning the surface.

15.The robot of claim 14, further comprises at least one tool holder holding a respective tool to which the tool head is attachable, the tool and tool head together forming an end effector, the robotic arm being configured to connect to a said tool such that using the tool head comprises controlling the tool.

16.The robot of claim 15, wherein the robotic arm and tool couple using a pneumatic coupling.

17.The robot of any one of claims 14 to 16, wherein the tool head disposal unit comprises a tool head remover and delivering the tool head comprises removing the tool head using the tool head remover.

18.The robot of claim 17, wherein the tool head remover aligns the tool head with a disposal bin.

19.The robot of any one of claims 14 to 18, further comprising one or more utilities lines extending from a body of the robot to a distal end of the robotic arm.

20.The robot of claim 19, wherein the one or more utilities lines connect from the distal end to the tool head.

21.The robot of claim 15 or 16, further comprising a control system configured to perform at least one of:

aligning the autonomous cleaning robot with an object according to the method of any one of claims 1 to 7; and

manipulating the robotic arm according to the method of any one of claims 8 to 13.

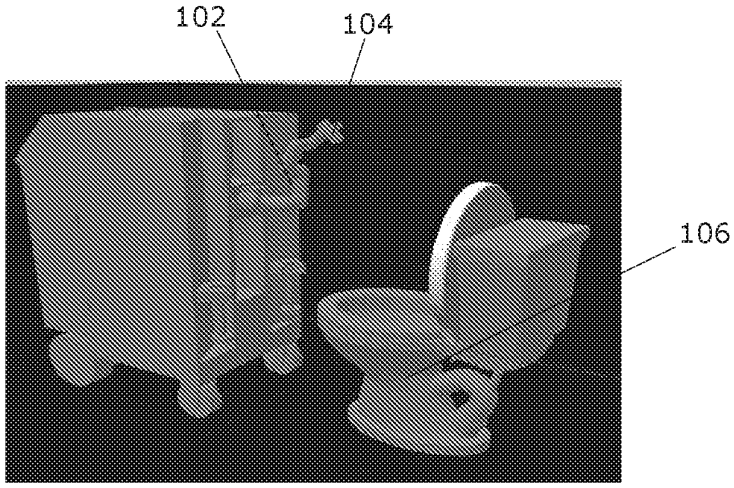


Figure 1

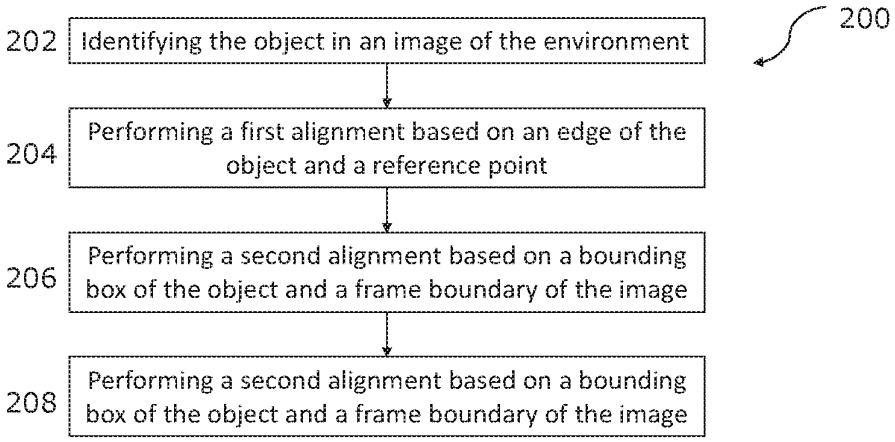

200
202
Identifying the object in an image of the environment
204
Performing a first alignment based on an edge of the object and a reference point
206
Performing a second alignment based on a bounding box of the object and a frame boundary of the image
208
Performing a second alignment based on a bounding box of the object and a frame boundary of the image

Figure 2

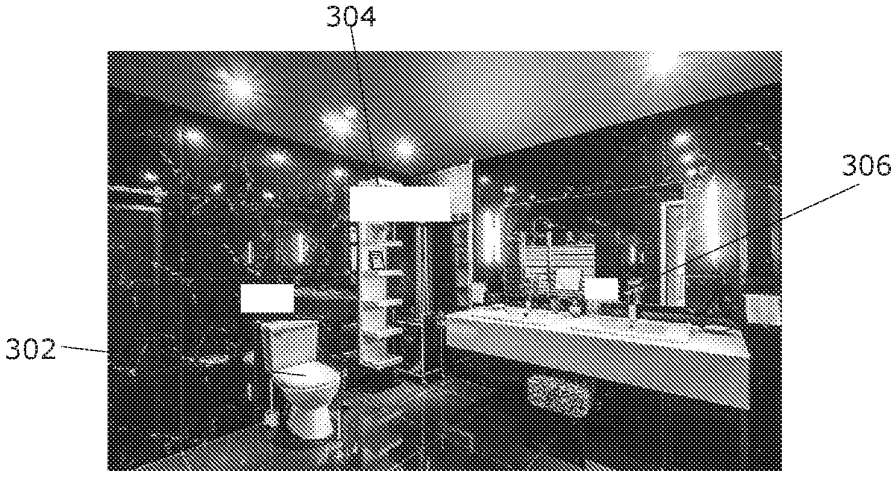


Figure 3

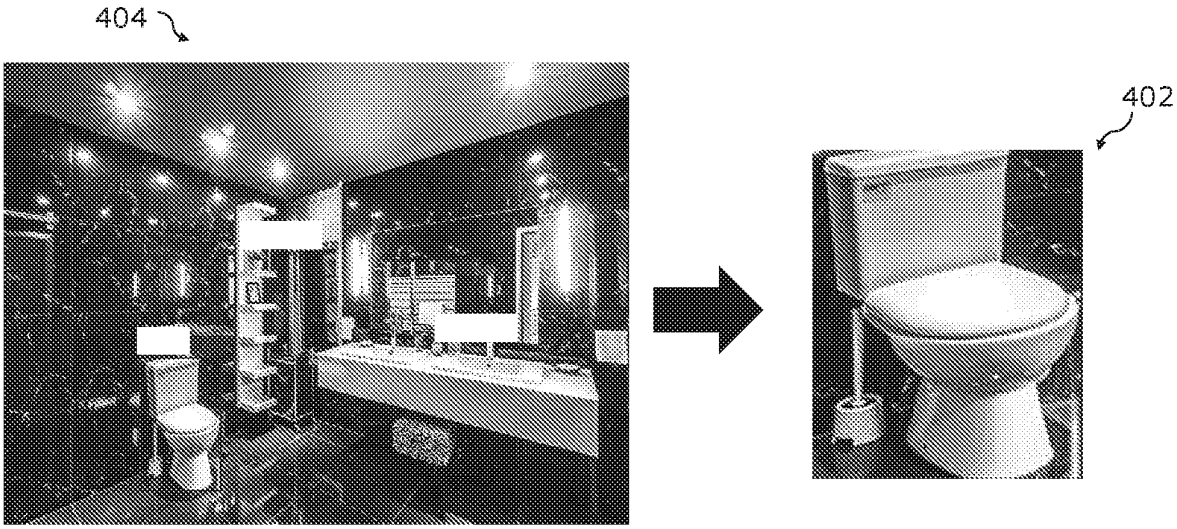


Figure 4

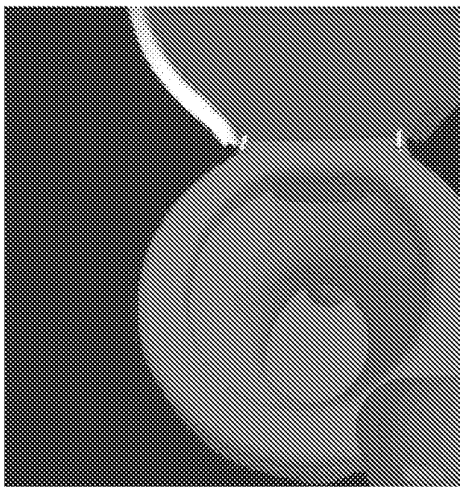


Figure 5

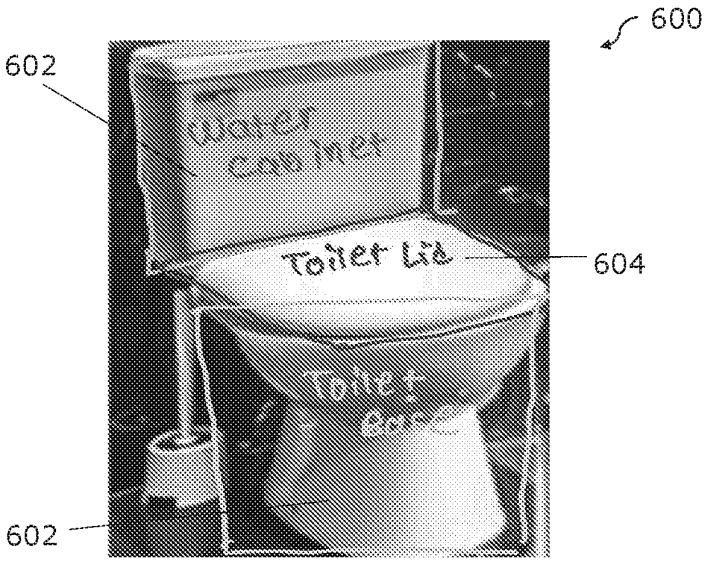


Figure 6



Figure 7A

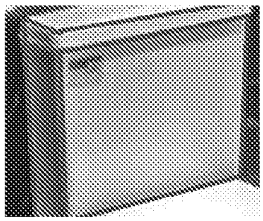


Figure 7B

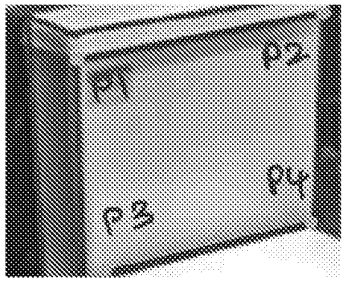


Figure 7C

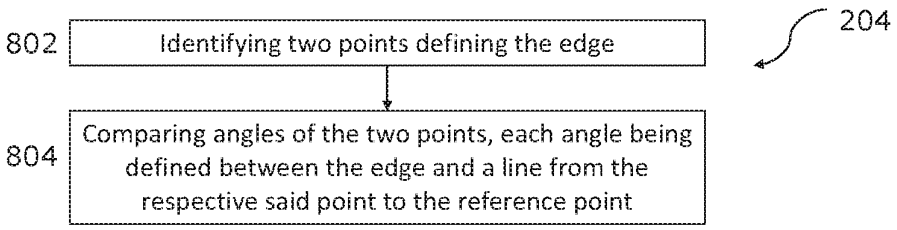

802
Identifying two points defining the edge
804
Comparing angles of the two points, each angle being defined between the edge and a line from the respective said point to the reference point
204

Figure 8

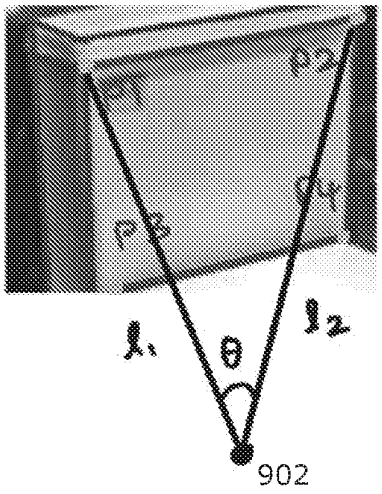


Figure 9

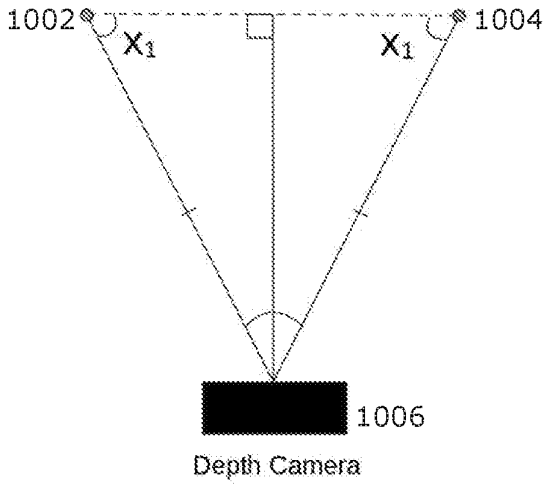


Figure 10

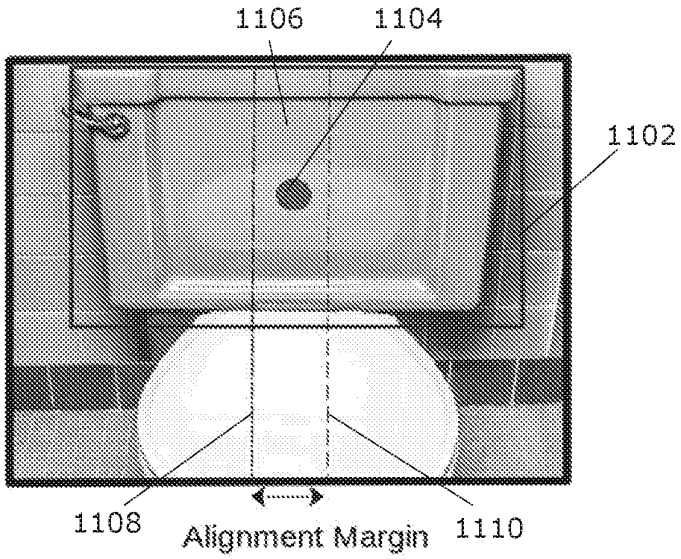


Figure 11

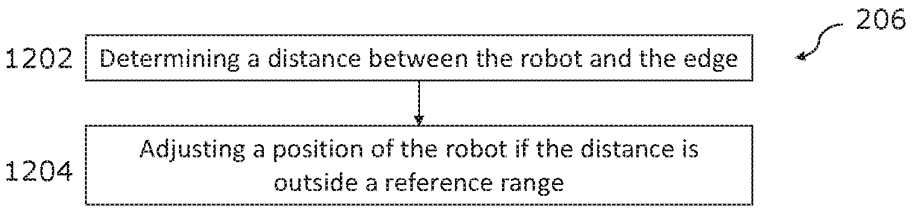

206
1202
Determining a distance between the robot and the edge
1204
Adjusting a position of the robot if the distance is outside a reference range

Figure 12

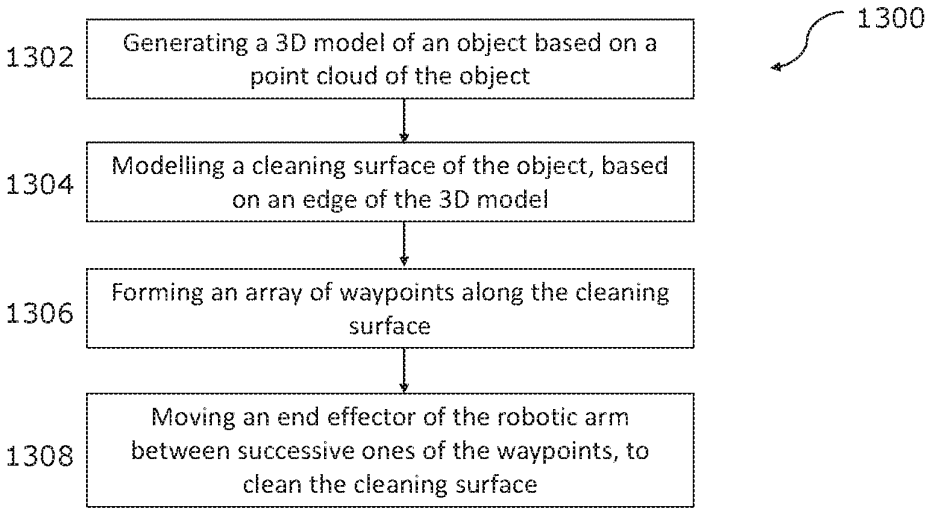

1300
1302
Generating a 3D model of an object based on a point cloud of the object
1304
Modelling a cleaning surface of the object, based on an edge of the 3D model
1306
Forming an array of waypoints along the cleaning surface
1308
Moving an end effector of the robotic arm between successive ones of the waypoints, to clean the cleaning surface

Figure 13

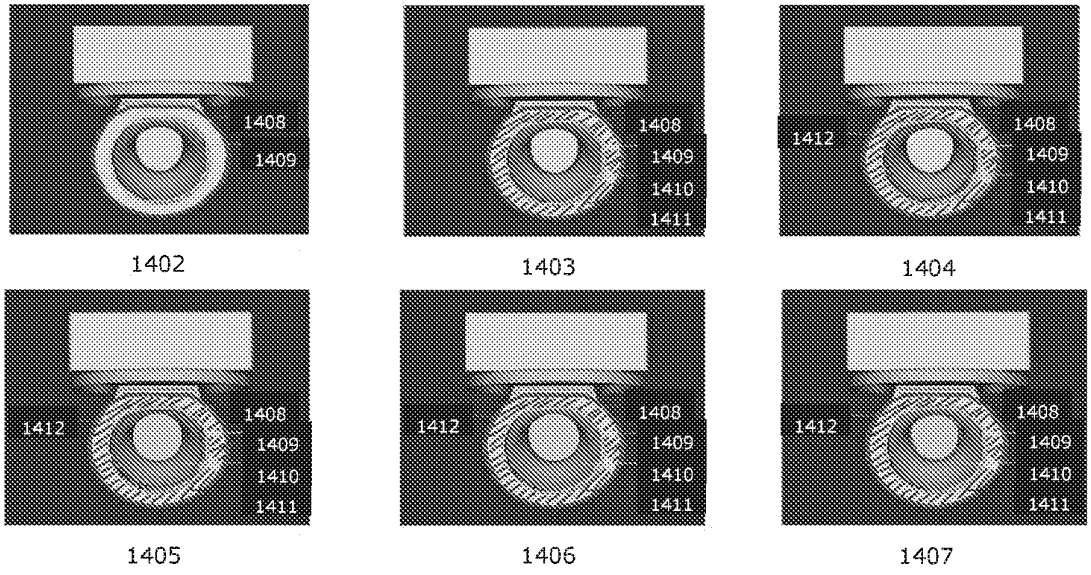

1408
1409
1402
1408
1409
1410
1411
1403
1412
1408
1409
1410
1411
1404
1412
1408
1409
1410
1411
1405
1412
1408
1409
1410
1411
1406
1412
1408
1409
1410
1411
1407

Figure 14

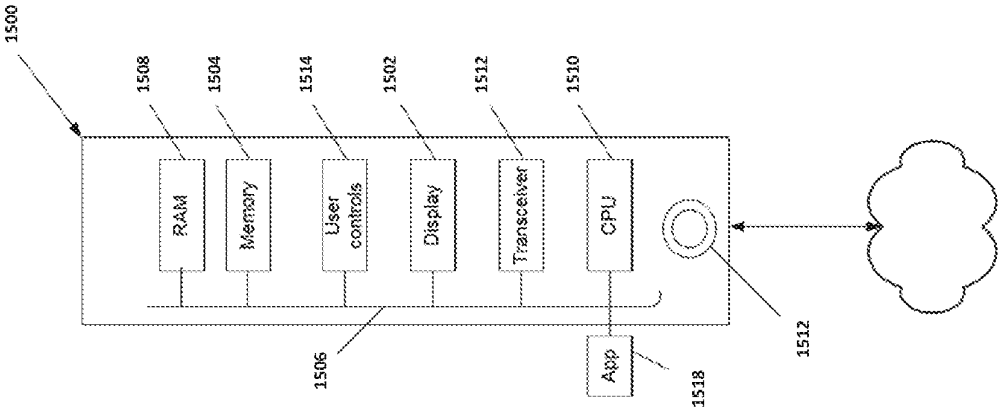

1500
1508
1504
1514
1502
1512
1510
RAM
Memory
User controls
Display
Transceiver
CPU
App
1506
1512
1518

Figure 15

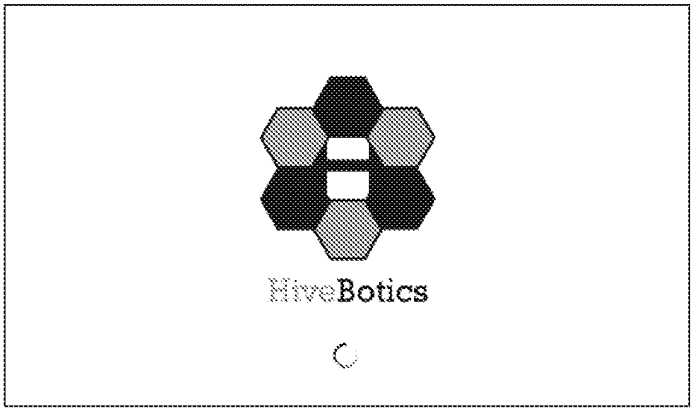


Figure 16

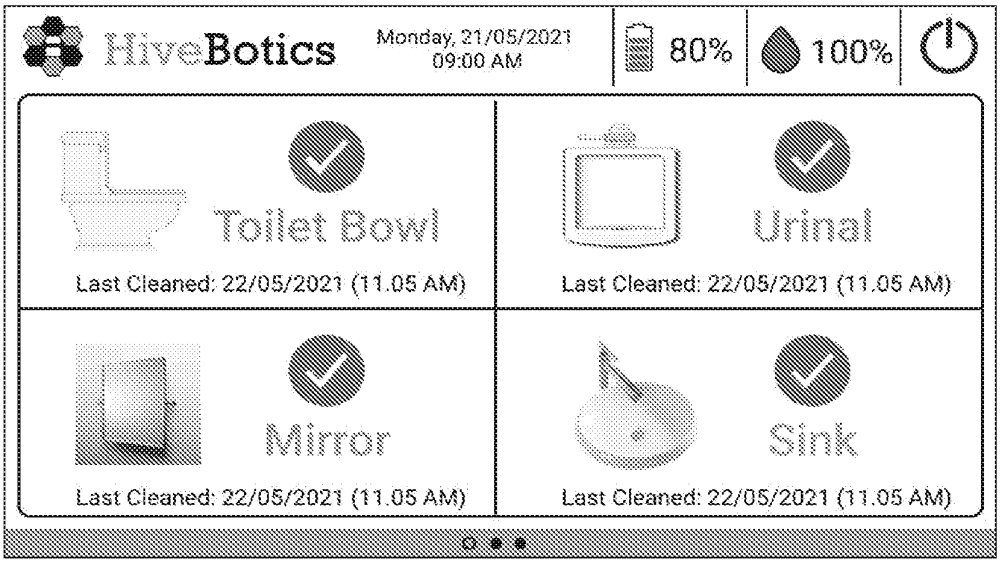

HiveBotics
Monday, 21/05/2021
09:00 AM
80%
100%
Toilet Bowl
Last Cleaned: 22/05/2021 (11.05 AM)
Urinal
Last Cleaned: 22/05/2021 (11.05 AM)
Mirror
Last Cleaned: 22/05/2021 (11.05 AM)
Sink
Last Cleaned: 22/05/2021 (11.05 AM)

Figure 17

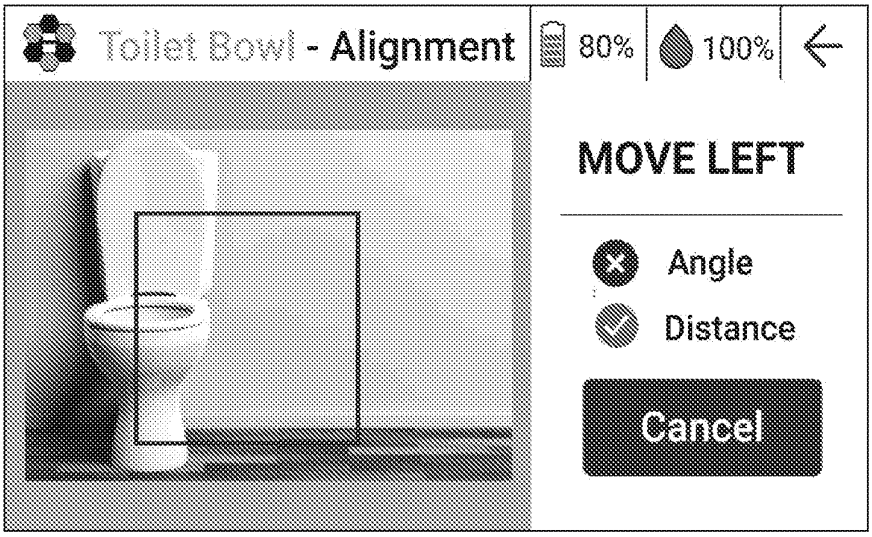

Toilet Bowl - Alignment
80%
100%
MOVE LEFT
Angle
Distance
Cancel

Figure 18

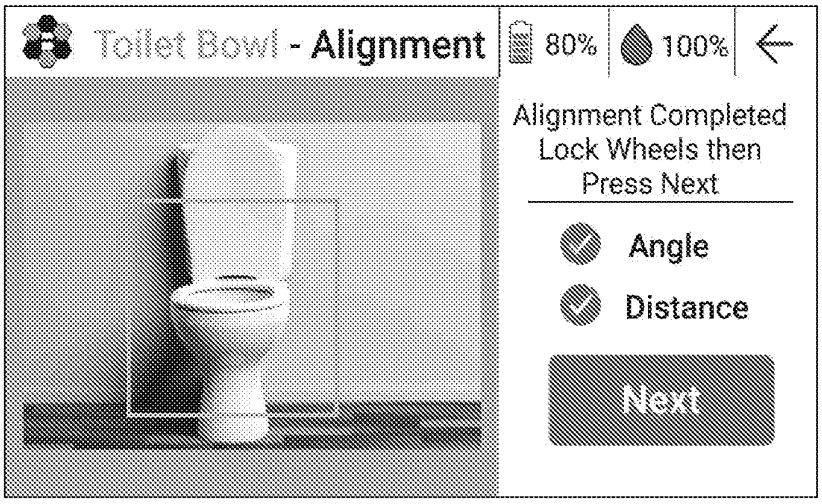

Toilet Bowl - Alignment
80%
100%
Alignment Completed
Lock Wheels then
Press Next
Angle
Distance
Next

Figure 19

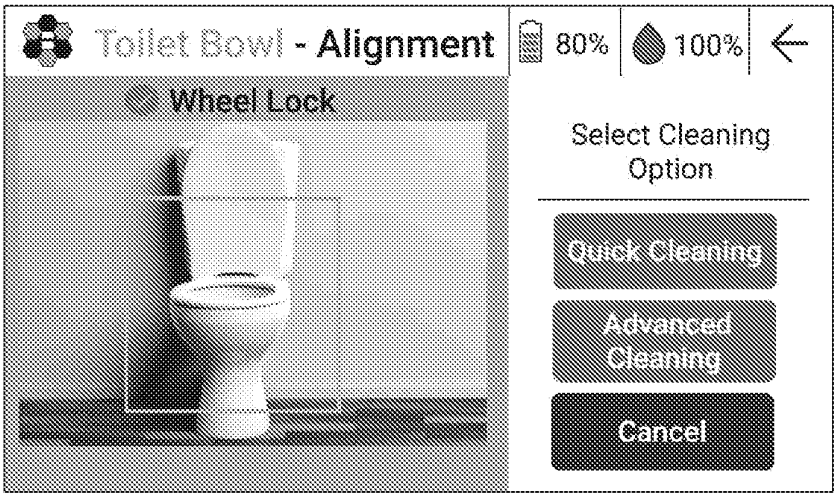

Toilet Bowl - Alignment
80%
100%
Wheel Lock
Select Cleaning Option
Quick Cleaning
Advanced Cleaning
Cancel

Figure 20

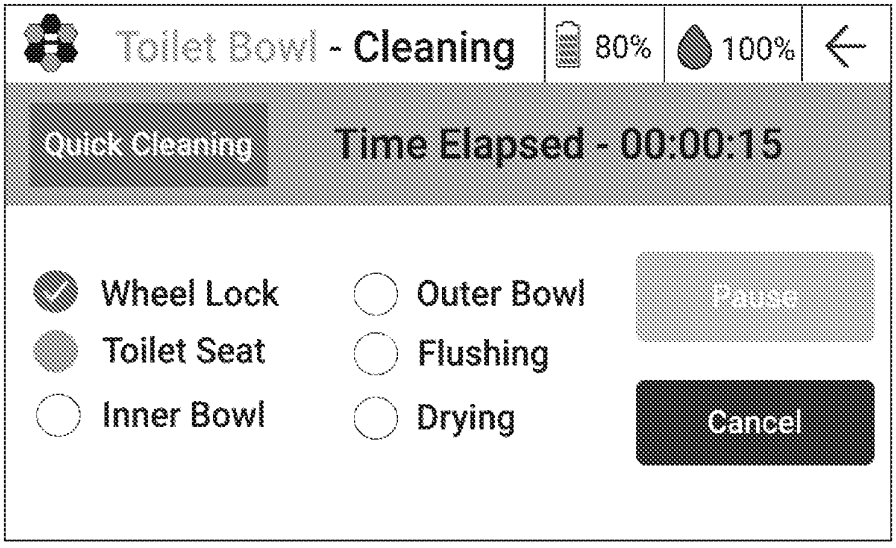

Toilet Bowl - Cleaning
80%
100%
Quick Cleaning
Time Elapsed - 00:00:15
Wheel Lock
Toilet Seat
Inner Bowl
Outer Bowl
Flushing
Drying
Pause
Cancel

Figure 21

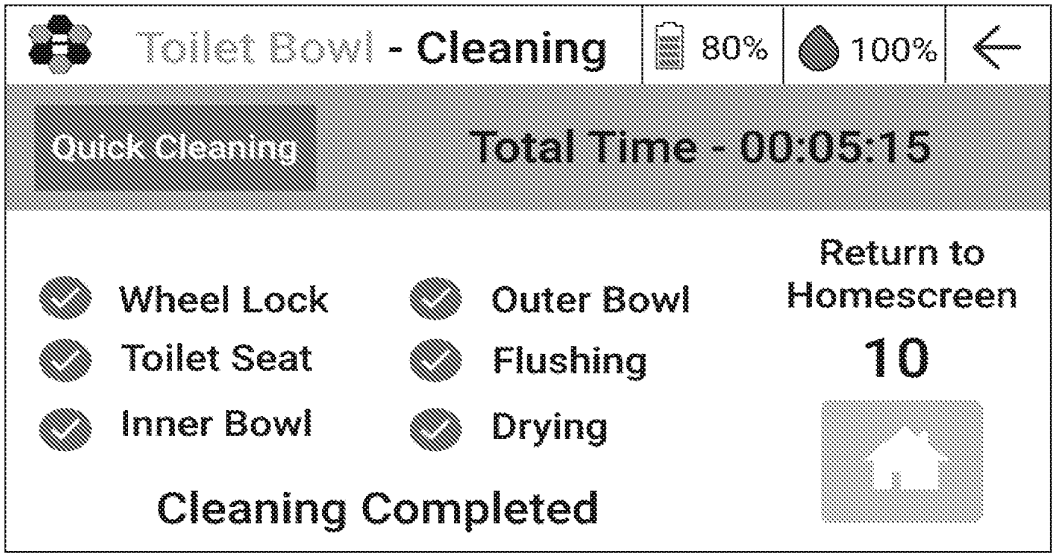

Toilet Bowl - Cleaning
80%
100%
Quick Cleaning
Total Time - 00:05:15
Wheel Lock
Outer Bowl
Toilet Seat
Flushing
Inner Bowl
Drying
Return to Homescreen
10
Cleaning Completed

Figure 22

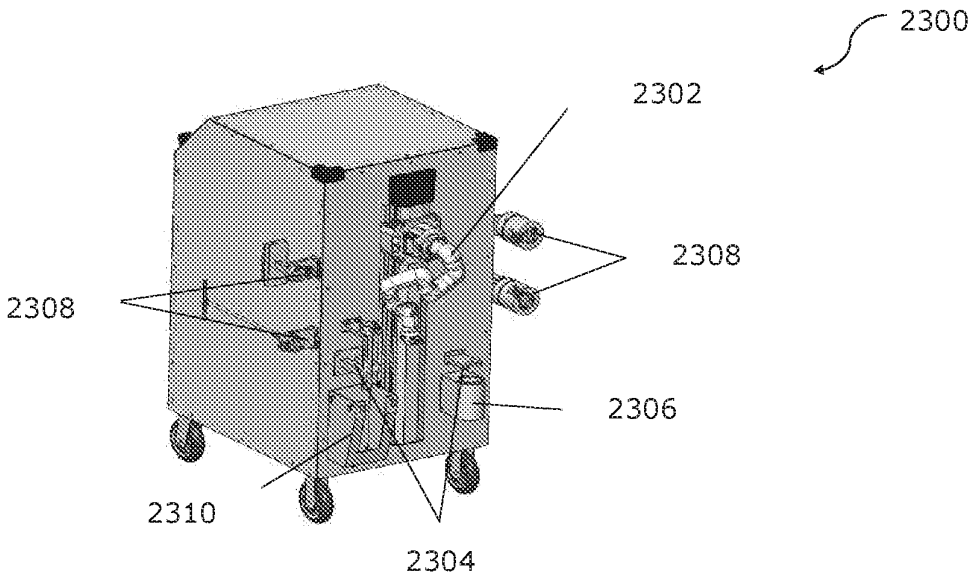


Figure 23

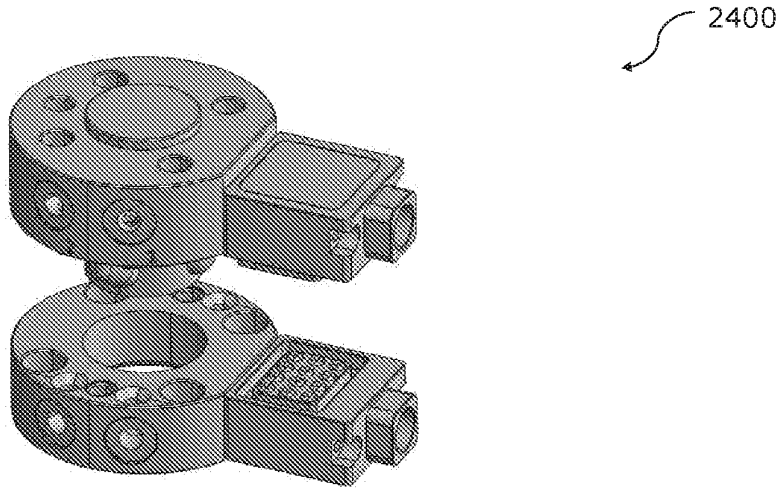


Figure 24

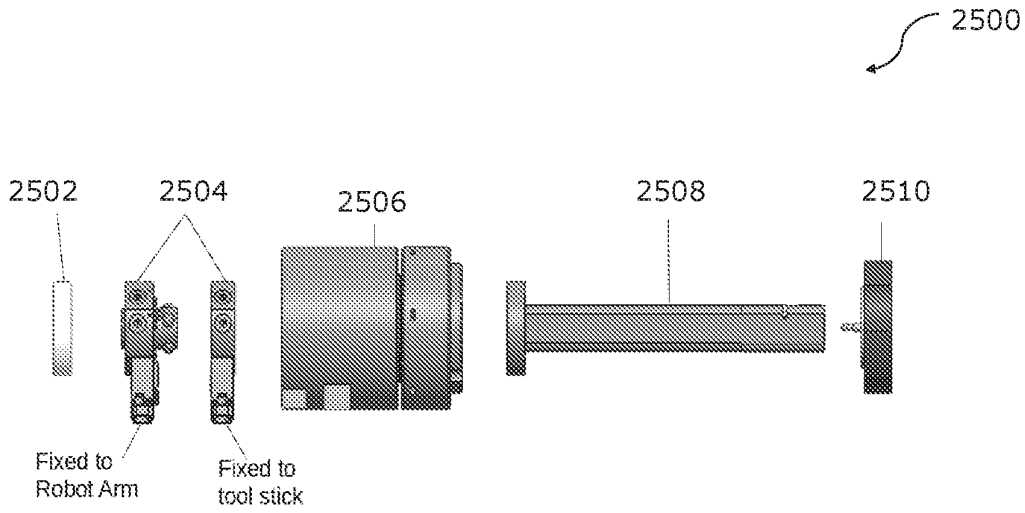

2500
2502
2504
2506
2508
2510
Fixed to
Robot Arm
Fixed to
tool stick

Figure 25

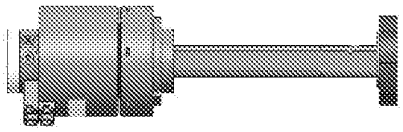


Figure 26A

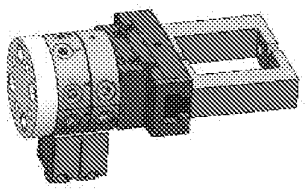


Figure 26B

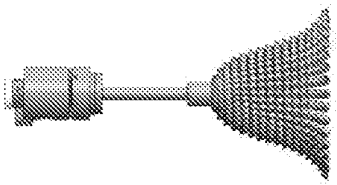


Figure 26C

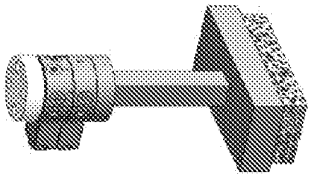


Figure 26D

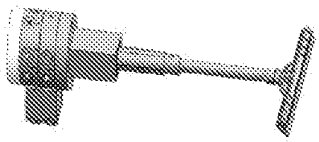


Figure 26E

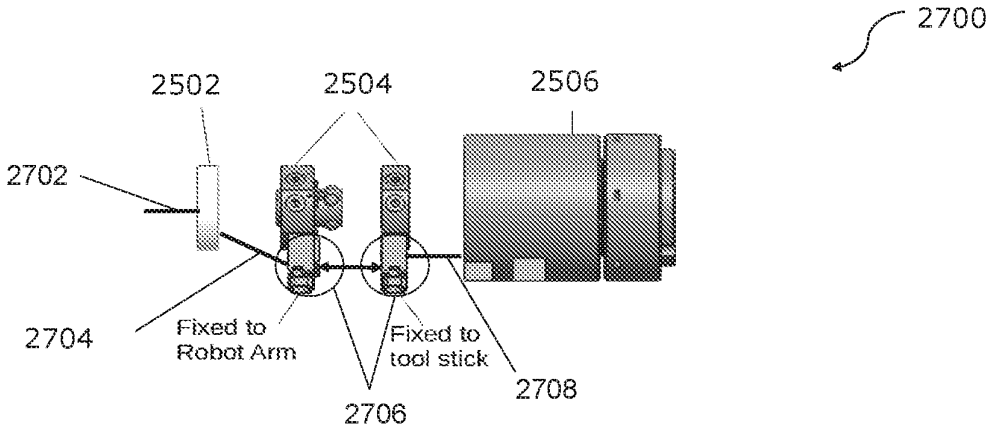

2700
2502
2504
2506
2702
2704
Fixed to
Robot Arm
Fixed to
tool stick
2706
2708

Figure 27

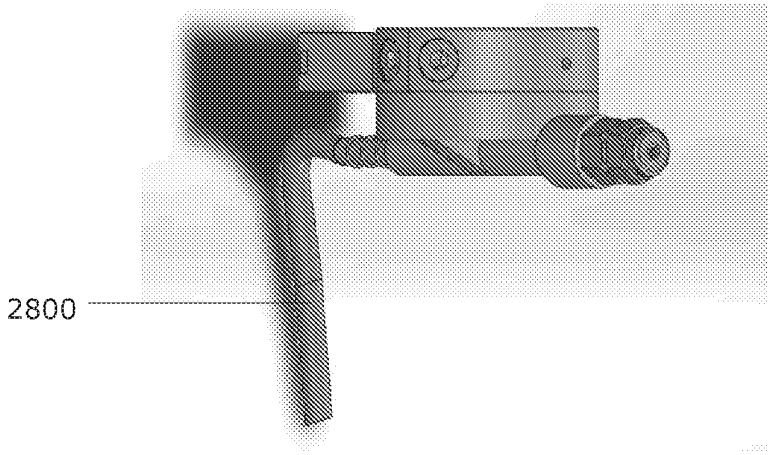


Figure 28

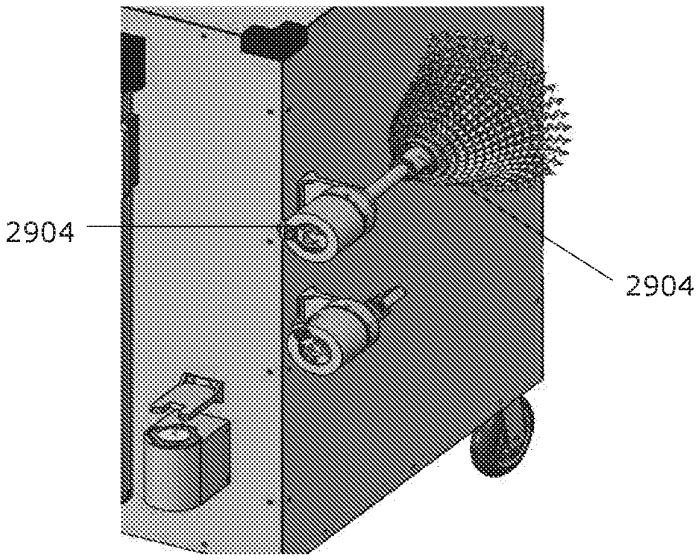


Figure 29

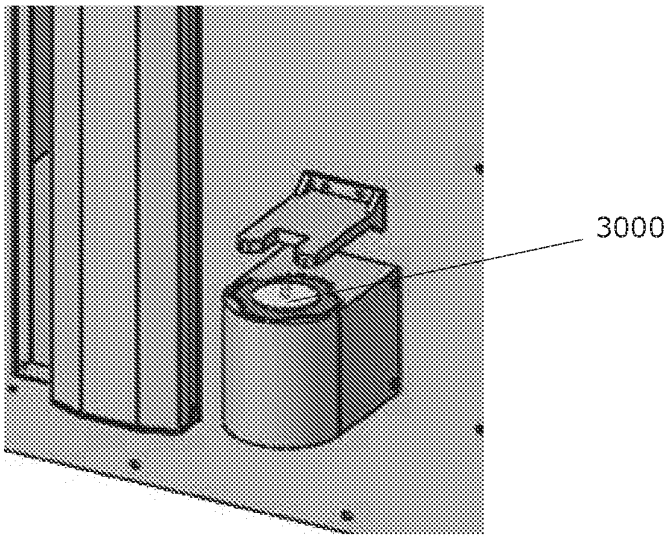


Figure 30

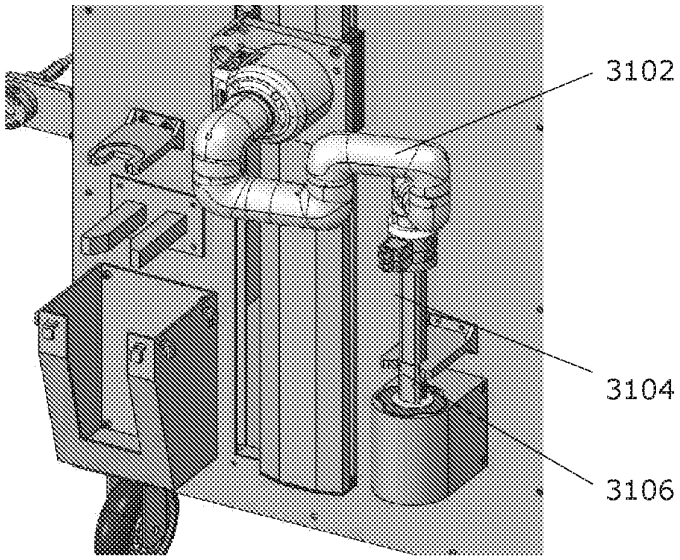


Figure 31

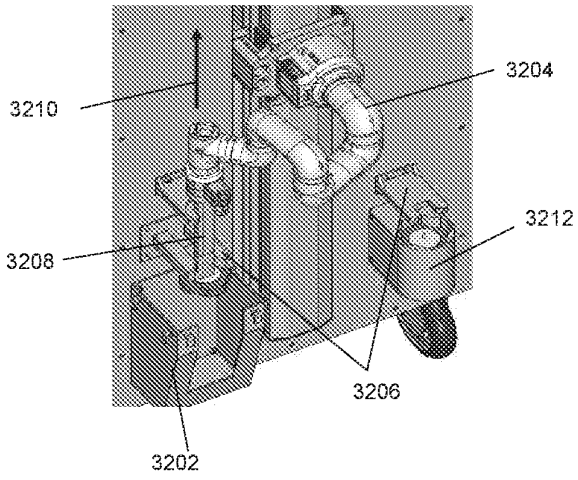


Figure 32

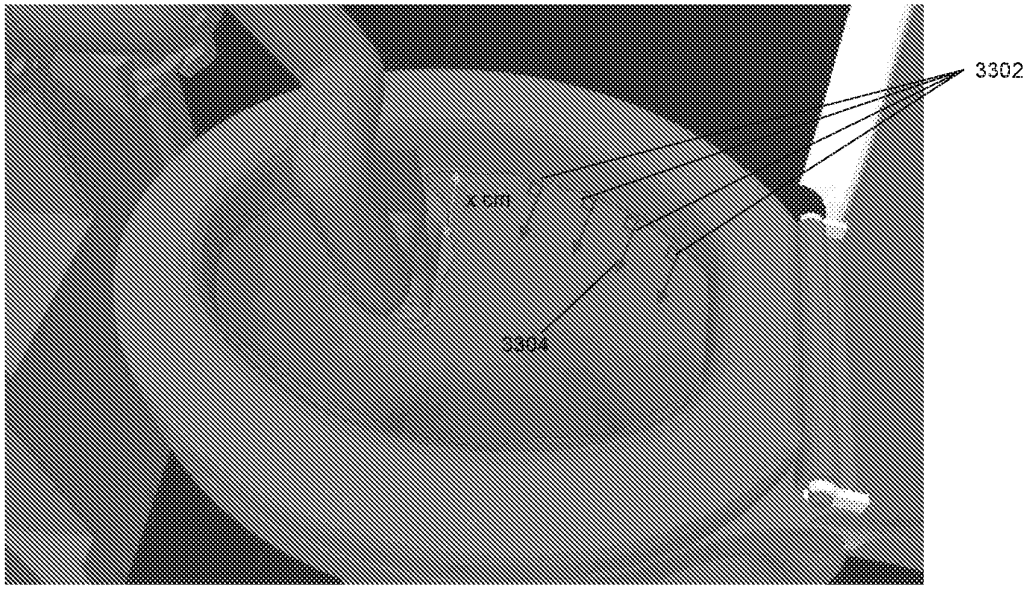


Figure 33

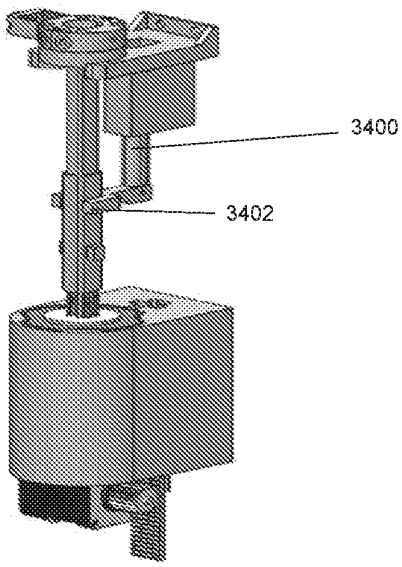


Figure 34

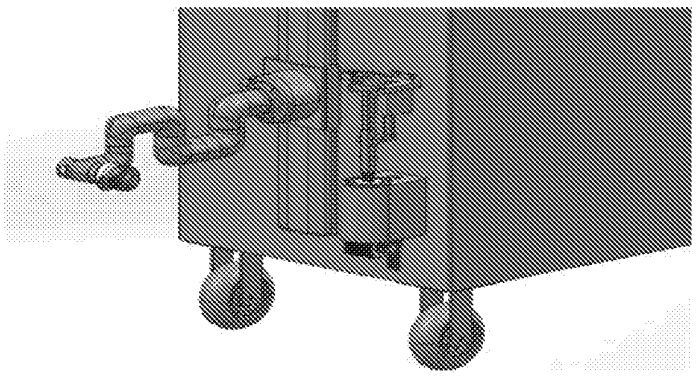


Figure 35